

**Minimum Fixed-Window Null Curvature Action Associated with
Hard Gyroscopic Gravitational-Wave Memory: A Sharp
Bel–Robinson Bound**

Susumu Watanuki

Independent Researcher, Chiba, Japan

Abstract

We derive a sharp lower bound linking the hard/helicity component of gyroscopic gravitational-wave memory generated by weak radiation to a Bel–Robinson curvature action measured on a fixed auxiliary null Jacobi probe crossing the same local radiation-zone data. The result is restricted to four dimensions, leading radiation-zone order, the standard initially shear-free Bondi frame, and linearized TT/Bondi radiative data that are stationary in the asymptotic past and future. The null Jacobi scattering scalar ω_γ and the gyroscope orientation angle $\Delta\Psi_{\text{H}}$ along a timelike worldline are distinct observables, but their quadratic functionals of the same local radiative data satisfy

$$\omega_\gamma^{(2)} = -2\Delta\Psi_{\text{H}}^{(2)}, \quad |\omega_\gamma^{(2)}| = 2|\Delta\Psi_{\text{H}}^{(2)}|.$$

Stationarity at the endpoints imposes mean-zero conditions on the two tidal-polarization controls. The physical minimum-action problem therefore reduces to the compressed spectral problem on

$$\mathcal{U}_0 = \left\{ f \in L^2(0, 1) : \int_0^1 f(t) dt = 0 \right\}, \quad K_{\text{stat}} = P_0 K_0 P_0,$$

with

$$(K_0 f)(t) = \frac{1}{2} \int_0^1 (t-s)|t-s|f(s) ds,$$

where P_0 is the orthogonal projection onto $\mathcal{U}_0$. The stationary Jacobi value function obeys

$$\kappa_{\text{stat}}(\omega) = \frac{|\omega|}{\|K_{\text{stat}}\|} + O(\omega^2),$$

and the coefficient is sharp by exact endpoint recovery. For a weak-amplitude family we define the second-order perturbative coefficients by

$$Q_\gamma(\varepsilon) = \varepsilon^2 Q_\gamma^{(2)} + O(\varepsilon^3), \quad \Delta\Psi_{\text{H}}(\varepsilon) = \varepsilon^2 \Delta\Psi_{\text{H}}^{(2)} + O(\varepsilon^3).$$

The corresponding leading-order radiative-data problem is a purely quadratic variational problem and is solved exactly by

$$\boxed{\kappa_{\text{H}}^{\text{lead}}(\psi) = \frac{2}{\|K_{\text{stat}}\|} |\psi|}.$$

Here $\psi = \Delta\Psi_{\text{H}}^{(2)}$ is the quadratic perturbative coefficient rather than the full finite-amplitude memory, and $\kappa_{\text{H}}^{\text{lead}}$ minimizes $Q_\gamma^{(2)}$. A Gauss–Legendre Nyström check gives

$$\|K_{\text{stat}}\| = 0.0281140680170\dots, \quad \frac{2}{\|K_{\text{stat}}\|} = 71.13876223\dots$$

These decimal values are numerical validation only; the theorem coefficient is defined exactly by the operator norm.

Here

$$Q_\gamma = L^3 \int_\gamma B(k, k, k, k) d\lambda$$

is a dimensionless windowed Bel–Robinson curvature action associated with a fixed radiation-crossing null segment γ . It is not an exposure measured along the gyroscope worldline, nor is it Bondi, Isaacson, radiated, or detector energy. It is invariant under affine reparametrizations of the same fixed segment, but changing the segment endpoints defines a different optimization problem. Likewise, changing the radiation-crossing null direction or the screen identification defines a different auxiliary-probe problem; Q_γ is not treated as a direction-independent intrinsic exposure. The main theorem minimizes the leading nonzero weak-amplitude coefficient $Q_\gamma^{(2)}$ for this fixed null window. The claims are restricted to the hard/helicity component, weak amplitude, leading radiation-zone order, and an initially shear-free frame; the final shear is not constrained to vanish, so a constant displacement-memory component is allowed. We do not claim a result for the full gyroscopic memory, spin memory, all BMS frames, or a finite-amplitude uniform theorem. Nor do we prove that every recovery profile constructed here admits a global embedding into a complete asymptotically flat vacuum Einstein spacetime.

I. INTRODUCTION

Gravitational-wave memory refers broadly to persistent effects that remain after a burst of gravitational radiation has passed. In addition to displacement memory, persistent observables based on spin memory, center-of-mass memory, holonomy, curve deviation, and rotating test bodies have been studied extensively [1–3, 5, 7]. The gyroscopic gravitational memory considered here is the orientation change acquired by a distant freely falling gyroscope after the passage of radiation. Its leading r^{-2} contribution splits into soft and hard parts. Previous work showed that the soft contribution coincides with spin memory, while the hard contribution is quadratic in the Bondi shear and can be interpreted in terms of local gravitational-wave helicity and as the generator of local electric-magnetic duality on radiative phase space [4, 6]. We therefore do not claim the discovery of gyroscopic memory or helicity itself.

Our question is inverse. Given radiative data with a prescribed hard gyroscopic memory, how small can a Bel–Robinson curvature action be on a fixed null window crossing that same

waveform? The resource Q_γ used here is not an action along the gyroscope worldline. It is the integral of a fourfold null contraction of the Bel–Robinson tensor along an auxiliary radiation-crossing null-geodesic segment γ . For the exact Jacobi problem we use Q_γ itself, while for comparison with Bondi/TT radiative data we use its quadratic weak-amplitude coefficient $Q_\gamma^{(2)}$. By “sharp” we mean not only that a lower-bound coefficient exists, but that no larger universal coefficient can hold and that admissible recovery families can approach the bound arbitrarily closely.

The mathematical core is the compact skew-adjoint operator

$$(K_0 f)(t) = \frac{1}{2} \int_0^1 (t-s)|t-s|f(s) ds$$

acting on the two-polarization trace-free null Jacobi profile. For the unrestricted nonlinear Jacobi memory channel, $\|K_0\|^{-1}$ is the exact optimal coefficient for sufficiently small targets. Physical stationarity in the asymptotic past and future, however, forces the tidal controls to have zero mean. Consequently, the optimal coefficient relevant to the hard gyroscopic-memory problem is controlled not by the norm of K_0 on the full L^2 space but by the compressed operator

$$K_{\text{stat}} = P_0 K_0 P_0$$

on the mean-zero subspace. We solve this constrained minimum-action problem analytically and combine it with the factor-of-two relation between the Jacobi channel and hard gyroscopic memory in the standard initially shear-free Bondi frame. This yields the exact sharp leading-order radiative-data law

$$\kappa_{\text{H}}^{\text{lead}}(\psi) = \frac{2}{\|K_{\text{stat}}\|} |\psi|$$

for the quadratic hard-memory coefficient at fixed $Q_\gamma^{(2)}$. The exact nonlinear Jacobi endpoint value function is a distinct small-target asymptotic statement with an $O(\omega^2)$ correction, and we keep the two results separate.

We delimit the relation to prior work as follows. Jacobi propagators, plane-wave memory tensors, the Bel–Robinson tensor, gyroscopic gravitational memory, the $N_{AB}\tilde{C}^{AB}$ hard functional, and the helicity/duality interpretation are established ingredients [2, 4, 6, 7]. The result derived here is the constrained inverse optimization that connects the hard gyroscopic/helicity functional of the radiative data to the fixed-window Bel–Robinson curvature action of the same data and determines its best coefficient. Our final same-condition literature

search did not identify a prior theorem with the same simultaneous hypotheses, but we do not regard such a search as a proof of historical priority.

Once stationarity is imposed, the unrestricted coefficient $2/\|K_0\| = 21.96874242\dots$ is no longer optimal. Stationarity implies $\alpha, \beta \in \mathcal{U}_0$, and repeating both the exact endpoint expansion and the compact-operator recovery construction on this closed subspace gives the best coefficient $2/\|K_{\text{stat}}\|$. The unrestricted coefficient remains a valid weaker lower bound, but it is not sharp for the stationary class. The use of “sharp” in this paper refers to the restricted-operator theorem proved below.

II. GEOMETRIC AND PHYSICAL SETUP

A. Bondi frame and hard gyroscopic memory

Let (u, r, θ^A) be Bondi coordinates, with

$$g_{AB} = r^2 \gamma_{AB} + r C_{AB} + O(1), \quad N_{AB} = \partial_u C_{AB}.$$

As the reference for the hard/soft split, we fix the standard initially shear-free vacuum frame

$$C_{AB}(u = -\infty) = 0. \tag{1}$$

The final shear ΔC_{AB} may be nonzero, so ordinary displacement memory is not excluded.

Following the conventions of Seraj–Oblak and Faye–Seraj, define $\tilde{C}_{AB} = \epsilon_{CA} C^C{}_B$. The hard part of the gyroscope precession is

$$\hat{\Omega}_H = -\frac{1}{8} N_{AB} \tilde{C}^{AB}, \tag{2}$$

and the physical orientation angle in the radiation zone is

$$\Delta\Psi_H = \frac{1}{r^2} \bar{\Phi}_H + O(r^{-3}), \quad \bar{\Phi}_H = -\frac{1}{8} \int du N_{AB} \tilde{C}^{AB}. \tag{3}$$

We refer to $\Delta\Psi_H$ as the hard/helicity gyroscopic gravitational memory. It is neither the full gyroscopic memory nor the spin memory [4, 6].

B. Null Jacobi system

Consider an auxiliary null-geodesic segment $\gamma : [\lambda_0, \lambda_1] \rightarrow M$ crossing the radiation in a four-dimensional Lorentzian spacetime. Let λ be an affine parameter, $k = d\gamma/d\lambda$, and

$L = \lambda_1 - \lambda_0 > 0$. The segment γ is not the timelike gyroscope worldline. We also exclude an outgoing null generator comoving with the wave on $u = \text{const.}$; instead we choose a radiation-crossing null direction with $du/d\lambda \neq 0$ so that the leading radiative curvature is sampled. We further work in a local radiation-zone patch with L fixed and $L/r \rightarrow 0$ as $r \rightarrow \infty$, which pushes angular and radial drift effects along γ beyond leading order in $1/r$. The screen basis is adapted so that, in this local patch, its transverse components agree with the Bondi/TT polarization dyad. These are geometric hypotheses of the main theorem, not a direction-independent statement for arbitrary null directions at finite r .

At leading asymptotically flat, locally plane-wave order, $du/d\lambda$ is constant. Using positive affine freedom we set $du/d\lambda = 1$, so that the retarded-time width of the selected stationary-to-stationary window equals L . Let e_A be a parallel-propagated orthonormal screen basis and define

$$T_{AB}(\lambda) = -R(e_A, k, e_B, k). \quad (4)$$

If $\text{Ric}(k, k) = 0$, then T is a symmetric trace-free 2×2 matrix.

With $t = (\lambda - \lambda_0)/L$, define the dimensionless profile

$$A(t) = L^2 T(\lambda_0 + Lt).$$

The Jacobi fundamental matrix obeys

$$\frac{d\Phi}{dt} = \begin{pmatrix} 0 & I_2 \\ A(t) & 0 \end{pmatrix} \Phi, \quad \Phi(0) = I_4. \quad (5)$$

Writing the free-propagation matrix as $F_0(s) = \begin{pmatrix} I_2 & sI_2 \\ 0 & I_2 \end{pmatrix}$, define the scattering matrix with the flat drift removed by

$$S_\gamma = F_0(-1)\Phi(1). \quad (6)$$

For the block decomposition $S_\gamma = (S_{ij})$, define the real scalar ω_γ through

$$\text{skew}(S_{11} + S_{22}) = \omega_\gamma J_2, \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (7)$$

We call ω_γ the nonlinear Jacobi memory/scattering channel. In the plane-wave memory-tensor language, it corresponds to the antisymmetric combination of the diagonal Jacobi blocks [2, 7].

C. Bel–Robinson curvature action

We use metric signature $(-+++)$ and normalize the Bel–Robinson tensor as

$$B_{abcd} = \frac{1}{4} (C_{aecf} C_b{}^e{}_d{}^f + {}^*C_{aecf} {}^*C_b{}^e{}_d{}^f). \quad (8)$$

On a vacuum null screen,

$$B(k, k, k, k) = |\Psi_0|^2 = \frac{1}{2} \text{tr}(T^2). \quad (9)$$

We therefore define

$$Q_\gamma := L^3 \int_{\lambda_0}^{\lambda_1} B(k, k, k, k) d\lambda = \frac{1}{2} \int_0^1 \text{tr}(A(t)^2) dt. \quad (10)$$

Under a positive affine reparametrization $\lambda' = a\lambda + b$ with $a > 0$, one has $L' = aL$, $k' = a^{-1}k$, $d\lambda' = a d\lambda$, and $B(k', k', k', k') = a^{-4}B(k, k, k, k)$. Hence $L'^3 d\lambda' B(k'^4) = L^3 d\lambda B(k^4)$. Thus the normalization $du/d\lambda = 1$ merely selects a representative of the affine-equivalence class of the fixed radiation-crossing segment and does not spoil the affine invariance of Q_γ . Changing the segment endpoints, the radiation-crossing null direction, or the screen identification instead defines a different auxiliary-probe problem; Q_γ is not window independent or direction independent.

Remark 1. Q_γ is a dimensionless windowed null action quadratic in the Weyl curvature. It is not an exposure measured on the gyroscope worldline, the Bondi mass loss, Isaacson effective stress-energy, radiated energy, or detector energy.

III. JACOBI MEMORY AND THE MINIMUM-ACTION PROBLEM

Using the STF basis

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

write

$$A(t) = \alpha(t)\sigma_3 + \beta(t)\sigma_1, \quad Q_\gamma = \|\alpha\|_2^2 + \|\beta\|_2^2. \quad (11)$$

Definition 2. Define

$$(K_0 f)(t) = \frac{1}{2} \int_0^1 (t-s)|t-s|f(s) ds, \quad s_0 := \|K_0\|. \quad (12)$$

The kernel is antisymmetric, so $K_0^* = -K_0$; it is also Hilbert–Schmidt and hence compact.

Lemma 3 (Small-amplitude expansion of the exact endpoint map). *For sufficiently small A , there exists $C > 0$ such that*

$$\omega_\gamma[A] = 2\langle\alpha, K_0\beta\rangle + R_\omega[A], \quad |R_\omega[A]| \leq CQ_\gamma[A]^2. \quad (13)$$

Proof. Write each Jacobi column of Eq. (5) as $X'' = AX$. Then

$$X(t) = X(0) + tX'(0) + (\mathcal{V}_A X)(t), \quad (\mathcal{V}_A X)(t) := \int_0^t (t-s)A(s)X(s) ds.$$

Since the interval has unit length,

$$\|\mathcal{V}_A X\|_\infty \leq \|A\|_{L^1, F} \|X\|_\infty \leq \|A\|_{L^2, F} \|X\|_\infty.$$

Hence, for sufficiently small $\|A\|_{L^2, F}$, the Neumann–Volterra series converges absolutely and the terminal values $X(1)$ and $X'(1)$, and therefore $\Phi(1)$ and $\omega_\gamma[A]$, are analytic in A near the origin. Let

$$\mathcal{V} = \text{span}\{\sigma_1, \sigma_3\}, \quad \mathcal{E} = \text{span}\{I_2, J_2\}.$$

Then $\mathcal{V}\mathcal{V} \subset \mathcal{E}$ and $\mathcal{E}\mathcal{V} \subset \mathcal{V}$. Consequently, any product of an odd number of $A(t_i) \in \mathcal{V}$ lies again in the symmetric space $\mathcal{V}$ and cannot contribute to the antisymmetric channel. The linear term is also symmetric. In the second-order time-ordered term, only the noncommuting part of the two STF basis matrices survives. Using $[\sigma_3, \sigma_1] = -2J_2$ yields the kernel $\frac{1}{2}(t-s)|t-s|$ and therefore the leading term $2\langle\alpha, K_0\beta\rangle$. The next possible contribution is fourth order, so in a fixed neighborhood

$$|R_\omega[A]| \leq C_1 \|A\|_{L^1, F}^4 \leq C_1 \|A\|_{L^2, F}^4 = CQ_\gamma^2.$$

□

IV. SHARP MINIMUM-ACTION THEOREM

Define

$$\kappa_\Omega(\omega) := \inf\{Q_\gamma[A] : \omega_\gamma[A] = \omega\}. \quad (14)$$

Theorem 4 (Sharp law for the unrestricted nonlinear Jacobi channel).

$$\boxed{\kappa_\Omega(\omega) = \frac{|\omega|}{\|K_0\|} + O(\omega^2), \quad \omega \rightarrow 0.} \quad (15)$$

The coefficient is sharp for the exact endpoint problem.

Proof. By Cauchy–Schwarz and the arithmetic–geometric mean inequality,

$$2|\langle \alpha, K_0 \beta \rangle| \leq s_0(\|\alpha\|_2^2 + \|\beta\|_2^2) = s_0 Q_\gamma. \quad (16)$$

Compactness implies that the norm of K_0 is attained. Choose a real unit vector β_* in the top eigenspace of $-K_0^2$,

$$-K_0^2 \beta_* = s_0^2 \beta_*, \quad \alpha_* := K_0 \beta_* / s_0.$$

Then $\|\alpha_*\| = \|\beta_*\| = 1$ and $\langle \alpha_*, K_0 \beta_* \rangle = s_0$. For $A_* = \alpha_* \sigma_3 + \beta_* \sigma_1$, one has $Q_\gamma[A_*] = 2$ and a quadratic memory coefficient $2s_0$, saturating Eq. (16).

For exact recovery, set $A_\rho = \rho A_*$. The preceding lemma gives

$$\omega_\gamma[A_\rho] = 2s_0 \rho^2 + O(\rho^4), \quad Q_\gamma[A_\rho] = 2\rho^2. \quad (17)$$

Analyticity then yields

$$\frac{d}{d\rho} \omega_\gamma[A_\rho] = 4s_0 \rho + O(\rho^3) > 0$$

for sufficiently small $\rho > 0$. Thus we do not invoke the usual inverse-function theorem at $\rho = 0$, where the derivative vanishes; instead, strict one-sided monotonicity gives a unique $\rho(\omega) > 0$ for every sufficiently small positive target. Hence

$$Q_\gamma[A_{\rho(\omega)}] = \frac{\omega}{s_0} + O(\omega^2).$$

For negative targets replace β_* by $-\beta_*$; the same one-sided argument gives the negative branch.

Conversely, the recovery construction already implies $\kappa_\Omega(\omega) = O(|\omega|)$, so a minimizing sequence may be chosen with $Q_\gamma = O(|\omega|)$. Applying Eqs. (13) and (16) along that sequence gives

$$|\omega| \leq s_0 Q_\gamma + C Q_\gamma^2.$$

Since $Q_\gamma = O(|\omega|)$, this yields $Q_\gamma \geq |\omega|/s_0 - O(\omega^2)$. Combining the lower and upper bounds proves Eq. (15). $\square$

A Gauss–Legendre Nyström calculation gives

$$\|K_0\| = 0.09103843824\dots, \quad \|K_0\|^{-1} = 10.98437121\dots \quad (18)$$

These decimal values are numerical validation only and do not define the theorem constant.

V. RELATION TO GYROSCOPIC MEMORY

A. Polarization area of a weak TT wave

On the same local radiation-zone patch with the adapted screen fixed in the preceding section, write the leading radiative data at a large reference Bondi radius r as a weak TT strain

$$h^{TT}(u, r; \varepsilon) = \varepsilon \begin{pmatrix} p(u) & q(u) \\ q(u) & -p(u) \end{pmatrix} + O(\varepsilon^2). \quad (19)$$

In the radiation zone, $p, q = O(r^{-1})$. We work at fixed r and suppress the r argument below. Stationarity before and after the burst implies $\dot{p}(u_{\pm}) = \dot{q}(u_{\pm}) = 0$. If

$$X(\varepsilon) = X^{(0)} + \varepsilon X^{(1)} + \varepsilon^2 X^{(2)} + O(\varepsilon^3),$$

we denote the coefficient of ε^2 by $X^{(2)}$. Thus $\Delta\Psi_H^{(2)}$ is neither the full finite-amplitude memory nor the full order- ε^2 contribution with the explicit factor ε^2 included; it is the quadratic perturbative coefficient.

With metric signature $(-+++)$, the convention $T_{AB} = -R(e_A, k, e_B, k)$ from Eq. (4), and our outgoing/retarded-time convention, a radiation-crossing null line normalized by $du/d\lambda = 1$ and equipped with a screen adapted to the Bondi/TT polarization dyad has leading type-N radiative curvature

$$R(e_A, k, e_B, k) = -\frac{1}{2}\ddot{h}_{AB}^{TT} + (\text{subleading in } 1/r \text{ and nonlinear amplitude}).$$

Hence the leading tidal profile is $T = \frac{1}{2}\ddot{h}^{TT}$, so the same pair (p, q) can be inserted into both the Jacobi and Bondi/TT descriptions. The required statement is this fixed local identification; we do not assert that arbitrary null directions at finite r give the same T . The overall sign of the TT/Bondi relation is convention dependent, but a simultaneous sign flip of (p, q) leaves the quadratic area below unchanged. We fix the sign convention in Eq. (23) and use it consistently.

The quadratic Jacobi coefficient is the signed polarization-plane area obtained by closing the trajectory (p, q) with a straight chord,

$$\mathcal{A}_{\text{pol}}[p, q] = \frac{1}{2} \int_{u_-}^{u_+} (p\dot{q} - q\dot{p}) du + \frac{1}{2}[p_f q_i - p_i q_f]. \quad (20)$$

More precisely,

$$\omega_\gamma(\varepsilon) = \varepsilon^2 \omega_\gamma^{(2)} + O(\varepsilon^3), \quad \omega_\gamma^{(2)} = \mathcal{A}_{\text{pol}}[p, q]. \quad (21)$$

In the initially shear-free reference frame, $p_i = q_i = 0$, so the chord term vanishes, while the final shear may remain nonzero.

B. Factor-of-two relation to the Bondi hard functional

On the local screen write

$$C = \begin{pmatrix} X & Y \\ Y & -X \end{pmatrix}, \quad \tilde{C} = \begin{pmatrix} -Y & X \\ X & Y \end{pmatrix}.$$

We use $\epsilon_{12} = +1$ and the same index ordering as Seraj–Oblak and Faye–Seraj, $\tilde{C}_{AB} = \epsilon_{CA} C^C{}_B$. Reversing the index order in the dual convention would flip the overall sign, but that is not the convention used here. Therefore

$$N_{AB} \tilde{C}^{AB} = 2(X\dot{Y} - Y\dot{X}). \quad (22)$$

Following the polarization convention of Faye–Seraj, we take in the radiation zone

$$C_{AB} = -r h_{AB}^{TT} + O(r^0). \quad (23)$$

Eqs. (3), (19), and (22) give

$$\Delta \Psi_{\text{H}}(\varepsilon) = \varepsilon^2 \Delta \Psi_{\text{H}}^{(2)} + O(\varepsilon^3), \quad \Delta \Psi_{\text{H}}^{(2)} = -\frac{1}{4} \int (p\dot{q} - q\dot{p}) du. \quad (24)$$

Under the initially shear-free condition, Eq. (21) becomes

$$\omega_\gamma^{(2)} = \frac{1}{2} \int (p\dot{q} - q\dot{p}) du. \quad (25)$$

Hence:

Theorem 5 (Factor relation between the two quadratic functionals of the same radiative data). *In the standard initially shear-free Bondi frame, at weak amplitude and leading radiation-zone order, and with the oriented-screen conventions above,*

$$\boxed{\omega_\gamma^{(2)} = -2\Delta \Psi_{\text{H}}^{(2)}}, \quad \boxed{|\omega_\gamma^{(2)}| = 2|\Delta \Psi_{\text{H}}^{(2)}|}. \quad (26)$$

The quantity $\omega_\gamma^{(2)}$ is a Jacobi scattering observable on a null geodesic, whereas $\Delta\Psi_H^{(2)}$ is an orientation angle measured on a timelike gyroscope worldline; they are not identified as the same observable. The equality states that, under the local radiation-zone patch, radiation-crossing null direction, and adapted-screen identification fixed above, the two known quadratic functionals evaluated on the same local TT/Bondi radiative data (p, q) differ by the factor shown. A different null direction or screen identification defines a new auxiliary-probe problem for which the relation must be rechecked. Reversing screen orientation flips both pseudoscalars simultaneously. We do not claim a theorem that controls finite r and finite amplitude simultaneously; the safe coefficient statement is the iterated limit

$$\lim_{r \rightarrow \infty} r^2 \lim_{\varepsilon \rightarrow 0} \varepsilon^{-2} (\omega_\gamma(\varepsilon, r) + 2\Delta\Psi_H(\varepsilon, r)) = 0. \quad (27)$$

VI. STATIONARY PHYSICAL MINIMUM-ACTION THEOREM

Endpoint stationarity determines the optimal coefficient. On the normalized interval, write the first-order tidal profile as

$$A^{(1)}(t) = \alpha(t)\sigma_3 + \beta(t)\sigma_1, \quad \alpha = \frac{1}{2}p'', \quad \beta = \frac{1}{2}q''.$$

Then $A(t; \varepsilon) = \varepsilon A^{(1)}(t) + O(\varepsilon^2)$. At fixed r , $p, q = O(r^{-1})$, so $\alpha, \beta = O(r^{-1})$ and both $Q_\gamma^{(2)}$ and $\Delta\Psi_H^{(2)}$ scale as $O(r^{-2})$. Therefore no additional factor of r remains in the ratio or in the optimal coefficient. The Bel–Robinson curvature action has expansion

$$Q_\gamma(\varepsilon) = \varepsilon^2 Q_\gamma^{(2)} + O(\varepsilon^3), \quad Q_\gamma^{(2)} = \|\alpha\|_2^2 + \|\beta\|_2^2. \quad (28)$$

while $p'(0) = p'(1) = q'(0) = q'(1) = 0$ implies

$$\int_0^1 \alpha(t) dt = \int_0^1 \beta(t) dt = 0. \quad (29)$$

Introduce

$$\mathcal{U}_0 := \left\{ f \in L^2(0, 1) : \int_0^1 f(t) dt = 0 \right\}, \quad P_0 f = f - \int_0^1 f(s) ds, \quad (30)$$

and define

$$K_{\text{stat}} := P_0 K_0 P_0, \quad s_{\text{stat}} := \|K_{\text{stat}}\|_{\mathcal{U}_0 \rightarrow \mathcal{U}_0}. \quad (31)$$

The operator K_{stat} is compact and skew-adjoint on $\mathcal{U}_0$.

Define the stationary exact-Jacobi value function by

$$\kappa_{\text{stat}}(\omega) := \inf\{Q_\gamma[A] : \omega_\gamma[A] = \omega, \alpha, \beta \in \mathcal{U}_0\}. \quad (32)$$

Theorem 6 (Sharp law for the stationary Jacobi channel).

$$\boxed{\kappa_{\text{stat}}(\omega) = \frac{|\omega|}{s_{\text{stat}}} + O(\omega^2), \quad \omega \rightarrow 0.} \quad (33)$$

The coefficient s_{stat}^{-1} is sharp for the exact endpoint problem.

Proof. For $\alpha, \beta \in \mathcal{U}_0$,

$$\langle \alpha, K_0 \beta \rangle = \langle \alpha, P_0 K_0 P_0 \beta \rangle = \langle \alpha, K_{\text{stat}} \beta \rangle.$$

Hence

$$2|\langle \alpha, K_0 \beta \rangle| \leq s_{\text{stat}}(\|\alpha\|^2 + \|\beta\|^2) = s_{\text{stat}} Q_\gamma.$$

The remainder estimate from the unrestricted theorem is unchanged on the closed subspace, so the lower-bound argument for Theorem 4 applies with s_0 replaced by s_{stat} .

For sharpness, choose a real unit eigenvector β_* of the top eigenvalue s_{stat}^2 of $-K_{\text{stat}}^2$ on the real Hilbert space $\mathcal{U}_0$, and set

$$\alpha_* := K_{\text{stat}} \beta_* / s_{\text{stat}}.$$

Then

$$\|\alpha_*\| = \|\beta_*\| = 1, \quad \langle \alpha_*, K_{\text{stat}} \beta_* \rangle = s_{\text{stat}}.$$

With $A_* = \alpha_* \sigma_3 + \beta_* \sigma_1$ and $A_\rho = \rho A_*$, the mean-zero constraints are preserved for every ρ , and

$$Q_\gamma[A_\rho] = 2\rho^2, \quad \omega_\gamma[A_\rho] = 2s_{\text{stat}}\rho^2 + O(\rho^4).$$

The same one-sided local inversion as in Theorem 4, together with a sign flip of one polarization for negative targets, exactly recovers sufficiently small targets of either sign. The upper and lower asymptotics therefore agree, proving Eq. (33). $\square$

Lemma 7 (Realization by stationary TT strain). *For any $\alpha, \beta \in \mathcal{U}_0$, define*

$$p(t) = 2 \int_0^t (t-s) \alpha(s) ds, \quad q(t) = 2 \int_0^t (t-s) \beta(s) ds. \quad (34)$$

Then

$$p(0) = q(0) = p'(0) = q'(0) = p'(1) = q'(1) = 0, \quad p'' = 2\alpha, \quad q'' = 2\beta.$$

Thus the controls lift to first-order TT data that are initially shear free and stationary both before and after the burst.

Proof. The initial conditions follow directly from the definition. Since $p'(t) = 2 \int_0^t \alpha(s) ds$, one has $p'(1) = 2 \int_0^1 \alpha = 0$, and similarly for q . The final values $p(1)$ and $q(1)$ are generally nonzero, so a constant final shear is allowed. $\square$

Lemma 8 (The same sharp constant for smooth stationary bursts). *Let*

$$\mathcal{U}_{0,c}^\infty := \left\{ f \in C_c^\infty(0,1) : \int_0^1 f(t) dt = 0 \right\}.$$

Then $\mathcal{U}_{0,c}^\infty$ is dense in $\mathcal{U}_0$. Consequently, restricting the admissible class to smooth stationary TT bursts whose curvature vanishes near the endpoints does not change the sharp leading-order coefficient s_{stat} .

Proof. For $f \in \mathcal{U}_0$, choose $g_n \in C_c^\infty(0,1)$ with $g_n \rightarrow f$ in L^2 . Then $\int g_n \rightarrow 0$. Fix $\chi \in C_c^\infty(0,1)$ with $\int \chi = 1$ and define

$$f_n := g_n - \left(\int_0^1 g_n \right) \chi.$$

Then $f_n \in \mathcal{U}_{0,c}^\infty$ and $f_n \rightarrow f$ in L^2 . Since K_{stat} is bounded, the bilinear form $\langle \alpha, K_{\text{stat}} \beta \rangle$ is continuous on $L^2 \times L^2$. Hence the top singular pair can be approximated arbitrarily well by smooth mean-zero functions. Each approximating pair vanishes near the endpoints in curvature, so the corresponding p, q from Eq. (34) extend smoothly as constants outside the active interval. The infimum over the smooth stationary-burst class therefore has the same sharp constant. $\square$

VII. MAIN THEOREM: SHARP LOWER BOUND ON BEL–ROBINSON CURVATURE ACTION

For a leading-order hard-memory target $\psi = \Delta \Psi_{\text{H}}^{(2)}$, define the value function of the *leading-order radiative-data problem* by

$$\kappa_{\text{H}}^{\text{lead}}(\psi) := \inf \left\{ Q_\gamma^{(2)}[A^{(1)}] : \Delta \Psi_{\text{H}}^{(2)}[A^{(1)}] = \psi, \alpha, \beta \in \mathcal{U}_0 \right\}. \quad (35)$$

The minimization is over the weak-amplitude TT/Bondi tangent-data class above; no value function over the full space of finite-amplitude Einstein solutions is defined.

Theorem 9 (Sharp minimum curvature action for a fixed radiation-crossing null window). *Assume the Bel–Robinson convention of Sec. II, a fixed radiation-crossing null window γ and adapted screen in a local radiation-zone patch with $L/r \rightarrow 0$, the standard initially shear-free Bondi frame, weak amplitude and leading radiation-zone order, and stationary-in/stationary-out conditions, without imposing zero final shear. Then*

$$\boxed{\kappa_{\text{H}}^{\text{lead}}(\psi) = \frac{2}{\|K_{\text{stat}}\|} |\psi|}. \quad (36)$$

This equality is exact for the leading-order perturbative radiative-data problem, and $2/\|K_{\text{stat}}\|$ is the optimal coefficient.

Proof. The quadratic leading term in Eq. (13), combined with Theorem 5, gives in the initially shear-free stationary class

$$\Delta\Psi_{\text{H}}^{(2)} = -\langle\alpha, K_{\text{stat}}\beta\rangle.$$

Therefore

$$2|\psi| = 2|\langle\alpha, K_{\text{stat}}\beta\rangle| \leq s_{\text{stat}}(\|\alpha\|_2^2 + \|\beta\|_2^2) = s_{\text{stat}}Q_{\gamma}^{(2)},$$

so $Q_{\gamma}^{(2)} \geq 2|\psi|/s_{\text{stat}}$.

Conversely, use the real top singular pair α_*, β_* constructed in the proof of Theorem 6. Choose the sign of β_* according to the sign of the target and set

$$\alpha = r\alpha_*, \quad \beta = \pm r\beta_*, \quad r^2 = \frac{|\psi|}{s_{\text{stat}}}.$$

Then $\Delta\Psi_{\text{H}}^{(2)} = \psi$ and

$$Q_{\gamma}^{(2)} = 2r^2 = \frac{2}{s_{\text{stat}}} |\psi|$$

exactly. Lemma 7 lifts the controls to initially shear-free, stationary-in/stationary-out TT data, and Lemma 8 shows that the same infimum is obtained after restricting to smooth bursts whose curvature vanishes near the endpoints. $\square$

Thus, for the fixed null window and within the stated leading-order class,

$$Q_{\gamma}^{(2)} \geq \frac{2}{\|K_{\text{stat}}\|} |\Delta\Psi_{\text{H}}^{(2)}|. \quad (37)$$

Numerically,

$$\|K_{\text{stat}}\| = 0.0281140680170\dots, \quad \boxed{\frac{2}{\|K_{\text{stat}}\|} = 71.13876223\dots} \quad (38)$$

These decimal values are validation only and do not define the theorem constant.

Remark 10 (Relation to the earlier coefficient $21.96874242\dots$). The unrestricted L^2 coefficient $2/\|K_0\| = 21.96874242\dots$ remains a valid weaker lower bound on the stationary subclass, but it is not sharp there. Under the mean-zero constraints the optimization must be redone for $K_{\text{stat}} = P_0 K_0 P_0$, yielding the optimal coefficient $2/\|K_{\text{stat}}\|$ above.

VIII. FRAME DEPENDENCE AND SCOPE

A. Why the initially shear-free frame is required

Under a time-independent shear shift

$$C_{AB} \mapsto C_{AB} + S_{AB},$$

the news is unchanged, but the hard functional changes as

$$\bar{\Phi}_H \mapsto \bar{\Phi}_H - \frac{1}{8} \Delta C_{AB} \tilde{S}^{AB}.$$

By contrast, the chord-closed area in Eq. (20) is invariant under $(p, q) \mapsto (p + p_0, q + q_0)$. Therefore the Jacobi area cannot in general be identified with the hard term alone in an arbitrary constant-shear reference. The initially shear-free frame is not merely a notational convenience; it is a physical convention required for the hard-component relation used here.

B. The fixed null window as auxiliary structure

The quantity Q_γ is not intrinsic to the waveform alone; it also depends on the chosen null window γ . For example, keep the curvature support and radiative history fixed, but append stationary curvature-free and news-free intervals at both ends so that the same $du/d\lambda = 1$ normalization extends the window length from L to $L' > L$. Then $\int_\gamma B(k^4) d\lambda$ and the hard memory remain unchanged, while

$$Q_{\gamma'} = \left(\frac{L'}{L}\right)^3 Q_\gamma.$$

Thus, for the same physical waveform, extending the physical window can make Q_γ arbitrarily large. The minimum-action law in this paper is therefore a dimensionless variational problem defined separately for each fixed window; it does not compare Q_γ values for different

windows as though they were a single detector exposure. This is distinct from an affine reparametrization of the same physical interval, under which Q_γ is invariant as shown in Sec. II.

C. Why stationarity changes the optimal coefficient

In the unrestricted problem, the top singular subspace of K_0 provides the recovery family. Stationary radiation, however, requires $\int \alpha = \int \beta = 0$. This imposes a codimension-one closed constraint on each polarization and cannot be removed by dense approximation. Numerically, the two constraints $\int \beta = 0$ and $\int K_0 \beta = 0$ are independent on the unrestricted top singular plane, so the old optimizer does not lie in the stationary class. The correct tangent-space operator is $K_{\text{stat}} = P_0 K_0 P_0$, and $\|K_{\text{stat}}\| < \|K_0\|$, so the sharp curvature-action coefficient increases.

We do not impose zero final shear. If one were also to require $p(1) = q(1) = 0$, then in addition to the mean-zero conditions one would have the first-moment constraints

$$\int_0^1 (1-t)\alpha(t) dt = 0, \quad \int_0^1 (1-t)\beta(t) dt = 0.$$

The admissible subspace and compressed operator would then differ from K_{stat} , and the best coefficient could change. The value $71.13876223\dots$ derived here must therefore not be transferred to that additional-constraint problem.

D. Claims not made

We do not claim:

- a sharp lower bound for the full gyroscopic gravitational memory;
- a sharp lower bound for spin memory;
- a theorem for the hard component alone that is invariant under arbitrary BMS-frame changes;
- a global finite-amplitude minimum-action law;
- that every recovery TT/Bondi datum constructed here admits a global realization by a complete asymptotically flat vacuum Einstein spacetime;

- an exact all-orders identity at finite r ;
- a lower bound on Bondi, Isaacson, radiated, or detector energy;
- the discovery of gyroscopic gravitational memory, the Bel–Robinson tensor, or Jacobi memory tensors themselves.

Accordingly, the “physical class” of Theorem 9 is the weak-amplitude, leading-radiation-zone TT/Bondi radiative-data class consistent with the standard initially shear-free convention, not the full global solution space of the Einstein equations. The quantities compared are the quadratic perturbative coefficients $Q_\gamma^{(2)}$ and $\Delta\Psi_H^{(2)}$, not the full finite-amplitude Q_γ and $\Delta\Psi_H$. Global nonlinear realizability and a finite-amplitude value function remain separate problems.

IX. NUMERICAL VALIDATION AND REPRODUCIBILITY

We distinguish analytic proof from numerical validation. The logic of Eqs. (15), (33), (26), and (36) is analytic. The calculations below validate normalization, sign, endpoint extraction, restricted spectral values, and counterexample-oriented test cases.

A. Unrestricted K_0 and exact Jacobi recovery

Gauss–Legendre Nyström discretization gives

nodes	$\ K_0\ $
50	0.0910384416041
100	0.0910384384527
200	0.0910384382516
300	0.0910384382407
500	0.0910384382384

Inserting the same discrete top singular pair into the exact Jacobi ODE gives

ρ	$Q_\gamma/ \omega_\gamma $
0.03	10.98435328
0.01	10.98436922
0.003	10.98437103
0.001	10.98437118

which converges to $\|K_0\|^{-1} = 10.98437121 \dots$. Here ρ is the amplitude along the recovery ray $A_\rho = \rho A_*$ and is distinct from the physical perturbation parameter ε . Because odd orders vanish in the antisymmetric Jacobi channel, $\omega_\gamma[A_\rho] = 2s_0\rho^2 + O(\rho^4)$, consistent with the observed $O(\rho^2)$ convergence of the ratio.

B. Stationary spectrum and exact recovery

Discretizing the mean-zero projection and computing the largest singular value of $P_0 M P_0$ gives

nodes	$\ K_{\text{stat}}\ $	$2/\ K_{\text{stat}}\ $
50	0.0281140528760	71.13880054294
100	0.0281140670547	71.13876466581
200	0.0281140679578	71.13876238062
300	0.0281140680065	71.13876225721
500	0.0281140680170	71.13876223067

A continuous reconstructed mean-zero top singular pair propagated through the exact Jacobi ODE gives

ρ	$Q_\gamma/ \omega_\gamma $
0.03	35.56942171300
0.01	35.56938562391
0.003	35.56938151880
0.001	35.56938115793

converging to $\|K_{\text{stat}}\|^{-1} = 35.5693811 \dots$. The ratio again has $O(\rho^2)$ error because of the even-order structure above. The reconstructed strain satisfies $p'(1) = q'(1) = 0$ to

numerical precision. With $\|\alpha_*\| = \|\beta_*\| = 1$, one orientation of the numerical SVD gives $p_f/\rho \simeq -0.47863$ and $q_f/\rho \simeq 0.30730$. The simultaneous sign-reversed pair represents the same optimizer because of the global sign ambiguity of the singular vectors. The final shear magnitude, independent of this global sign and of a polarization-basis rotation, is $\sqrt{p_f^2 + q_f^2}/\rho \simeq 0.56878$. The theorem therefore allows a constant displacement-memory component. These values check the implementation of the analytic recovery construction; they are not used to prove the restricted theorem.

C. Waveform checks of the bridge relation

Let $W = \int(p\dot{q} - q\dot{p})du$. In the initially zero reference, $\omega^{(2)} = W/2$ and $\Delta\Psi_H^{(2)} = -W/4$. Representative test families give

waveform family	W	$\omega^{(2)}$	$\Delta\Psi_H^{(2)}$	ratio
smooth circular polarization	2.35619447	1.17809724	-0.58904862	-2.000000
opposite helicity	-2.35619447	-1.17809724	0.58904862	-2.000000
elliptical polarization	1.22522112	0.61261056	-0.30630528	-2.000000
fixed axis	$\simeq 0$	$\simeq 0$	$\simeq 0$	both zero
endpoint memory	-0.62590250	-0.31295125	0.15647563	-2.000000
helicity cancellation	$\simeq 0$	$\simeq 0$	$\simeq 0$	both zero

As a counterexample-oriented baseline test, adding $(p_0, q_0) = (0.35, -0.22)$ to the endpoint-memory waveform leaves the chord-closed area unchanged, $-0.3129512514 \rightarrow -0.3129512515$, while the hard gyroscopic functional changes from 0.1564756257 to 0.1124756258. This falsifies a hard-component-only identity in an arbitrary constant-shear reference and supports the initially shear-free restriction.

D. Audit detecting the inadmissibility of the old optimizer

Choose an orthonormal basis of the unrestricted top singular subspace of K_0 . Normalize the two constraint rows associated with $\int \beta = 0$ and $\int K_0 \beta = 0$ by $\|K_0\|$. The absolute determinant of the resulting 2×2 matrix is invariant under orthogonal changes of basis and is approximately 0.90925 at $n = 500$. The same number equals the squared norm of the

orthogonal projection of the constant function onto the unrestricted top singular plane. Thus about 90.9% of the constant mode lies in that plane, and the two stationarity constraints are independent on it. This diagnoses why the unrestricted top singular subspace does not automatically satisfy stationarity. The sharpness of the new coefficient itself follows analytically from Theorem 6, not from this numerical audit.

X. PRIOR-ART BOUNDARY AND FORMULATION OF THE NOVELTY CLAIM

We classify the ingredients of the paper into established results, consequences derived directly from established results, and results derived here.

- Gyroscopic gravitational memory: established [4].
- Soft/spin and hard/helicity decomposition: established [4, 6].
- The $N_{AB}\tilde{C}^{AB}$ hard functional: established [4, 6].
- Helicity/local-duality-generator interpretation: established [4].
- Jacobi persistent observables and nonlinear plane waves: established [1, 2].
- Plane-wave memory tensors: established [7].
- Bondi–Sachs hierarchy of curve-deviation observables: established [3].
- Gyroscopic memory from holonomy: established [5].
- Bel–Robinson curvature algebra: established [8].
- Area-constrained curve optimization and bending energy: adjacent established variational problems [9, 10].
- Factor relation between polarization area and the hard gyroscopic functional: convention-fixed derivation from established functionals.
- Unrestricted Jacobi minimum-action law $1/\|K_0\|$: derived here.
- Stationary Jacobi minimum-action law $1/\|P_0K_0P_0\|$: derived here.
- Best coefficient $2/\|P_0K_0P_0\|$ for the fixed-window hard-memory problem: derived here.

Flanagan, Grant, Harte, and Nichols established the general framework of persistent observables and the Jacobi structure of nonlinear plane waves [1, 2]. Grant and Nichols developed curve-deviation observables in asymptotically flat spacetimes in the Bondi–Sachs framework [3]. Harte, Mieling, Oancea, and Steininger organized plane-wave scattering in terms of a finite set of memory tensors [7]. Seraj and Neogi related the Lorentz part of asymptotic holonomy to gyroscopic memory [5]. None of these structures is claimed as original here.

There are also neighboring mathematical problems. Hahmann, Sauvage, and Bonneau studied area-preserving multiresolution deformations of closed planar curves and associated bending-energy optimization [9]. Ferone, Kawohl, and Nitsch studied geometric elastica minimization under a fixed enclosed area [10]. These problems are close in spirit to the signed-area functional and second-derivative cost that appear here, but they differ in closed-curve/arc-length geometry, boundary conditions, the fixed-null-window Jacobi/Bondi pullback, and the mean-zero compressed operator. We therefore do not claim that area-constrained bending-energy optimization is new.

We make no claim of historical priority. The generic mechanisms of Cauchy–Schwarz, compact-operator norm saturation, and area-constrained curve optimization are established and are not counted as novelty. Our final same-condition literature search did not identify a prior theorem simultaneously imposing stationary-in/stationary-out conditions, an initially shear-free frame, a fixed radiation-crossing null window, a Bel–Robinson curvature action, a hard gyroscopic/helicity target, and an optimal coefficient. The novelty claimed here is restricted to the combination in which the physical stationarity conditions select the mean-zero space and the compressed operator $K_{\text{stat}} = P_0 K_0 P_0$, the fixed auxiliary-null-probe curvature action is related to the hard gyroscopic functional of the same radiative data, and the optimal coefficient is determined with exact Jacobi recovery. The literature search itself does not establish a historical first.

XI. DISCUSSION

The main result gives the best lower bound between an established hard gyroscopic functional and the leading perturbative coefficient of a curvature action measured on a fixed null window crossing the same radiative data. Once the quadratic hard-memory coefficient

is prescribed, no smaller $Q_\gamma^{(2)}$ can generate the same leading hard/helicity channel, while smooth stationary admissible recovery families approach the boundary arbitrarily closely. This is not merely a relabeling of the known helicity functional: it requires a constrained inverse optimization over waveform space. At the final mathematical step, however, the optimization is a standard operator-norm problem. The value of the result lies in the physical formulation, the constraint-selected operator, and the bridge between the two functionals under the stated conditions.

Stationarity substantially changes the coefficient. Formally transferring the unrestricted Jacobi coefficient to the hard-memory problem does not guarantee that the optimizing waveform is stationary before and after the burst. Imposing the physical conditions contracts the admissible tangent space to $\mathcal{U}_0$ and changes the relevant norm from $\|K_0\|$ to $\|P_0 K_0 P_0\|$. Consequently, the sharp coefficient is approximately 71.14, rather than approximately 21.97. This difference is a direct effect of the mean-zero compression.

The scope remains limited. First, Q_γ is not a standard gravitational-wave energy. Second, the hard gyroscopic memory is an r^{-2} effect and is quadratic in the weak-field radiative data, so it is not presently a directly detector-oriented observable. Third, the hard/soft split is frame dependent and the theorem is fixed to an initially shear-free convention. Fourth, the theorem concerns weak amplitude and leading radiation-zone order, not a global finite-amplitude value function on the nonlinear Einstein solution space.

The conclusions therefore have three layers. (i) The nonlinear Jacobi memory channel has an exact sharp small-target minimum-action law. (ii) With stationary-in/stationary-out conditions, the same structure survives on the mean-zero subspace through the compressed operator $K_{\text{stat}} = P_0 K_0 P_0$. (iii) In the standard initially shear-free Bondi radiation frame, the quadratic coefficient of that stationary Jacobi channel is minus twice the quadratic hard gyroscopic-memory coefficient, so the leading-order radiative-data problem with cost $Q_\gamma^{(2)}$ is solved exactly by $\kappa_{\text{H}}^{\text{lead}}(\psi) = 2|\psi|/\|K_{\text{stat}}\|$.

XII. CONCLUSION

We formulated a self-contained minimum-action problem for null Jacobi scattering and first proved, in the unrestricted class,

$$\kappa_{\Omega}(\omega) = \frac{|\omega|}{\|K_0\|} + O(\omega^2).$$

We then used the fact that physical stationarity imposes mean-zero tidal controls and obtained the restricted sharp law

$$\kappa_{\text{stat}}(\omega) = \frac{|\omega|}{\|K_{\text{stat}}\|} + O(\omega^2), \quad K_{\text{stat}} = P_0 K_0 P_0.$$

Finally, the local STF algebra of the Bondi shear and the polarization area of a weak TT wave give

$$\omega_{\gamma}^{(2)} = -2\Delta\Psi_{\text{H}}^{(2)}.$$

Therefore, in the standard initially shear-free frame, with stationary-in/stationary-out data at weak amplitude and leading radiation-zone order,

$$\kappa_{\text{H}}^{\text{lead}}(\psi) = \frac{2}{\|K_{\text{stat}}\|}|\psi|, \quad \frac{2}{\|K_{\text{stat}}\|} = 71.13876223\dots$$

Here $\psi = \Delta\Psi_{\text{H}}^{(2)}$ is a quadratic perturbative coefficient and the minimized cost is $Q_{\gamma}^{(2)}$. The coefficient is exactly sharp for the leading-order TT/Bondi radiative-data problem. The exact nonlinear Jacobi endpoint value function is instead governed by the distinct asymptotic theorem $\kappa_{\text{stat}}(\omega) = |\omega|/\|K_{\text{stat}}\| + O(\omega^2)$. A finite-amplitude hard-memory minimum-action law on the full Einstein solution space lies outside the claims of this paper.

Natural extensions include removing or replacing the auxiliary fixed-window structure, extending the value function to finite-amplitude Einstein solutions, and determining the best coefficient when the final shear is constrained as well.

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DATA AVAILABILITY STATEMENT

The custom code and numerical outputs supporting the numerical validation are publicly available in the companion repository at <https://github.com/stitch224/gravitational-memory-public> under `paper/01-hard-gyroscopic-memory/reproducibility`.

Appendix A: Convention ledger

- Metric signature: $(-+++)$.
- Bondi news: $N_{AB} = \partial_u C_{AB}$.
- Initial frame: $C_{AB}(-\infty) = 0$.
- Screen orientation: $\epsilon_{12} = +1$.
- Dual shear: $\tilde{C}_{AB} = \epsilon_{CA} C^C{}_B$.
- TT/Bondi relation: $C_{AB} = -r h_{AB}^{TT} + O(r^0)$ with the polarization convention fixed as in the main text.
- Optical tidal matrix: $T_{AB} = -R(e_A, k, e_B, k)$.
- Bel–Robinson tensor: Eq. (8), including the explicit factor $1/4$.
- Curvature action: $Q_\gamma = L^3 \int B(k^4) d\lambda$; for the representative $du/d\lambda = 1$, one has $d\lambda = du$.
- Leading physical cost: $Q_\gamma(\varepsilon) = \varepsilon^2 Q_\gamma^{(2)} + O(\varepsilon^3)$.
- Nonlinear Jacobi scalar: $\text{skew}(S_{11} + S_{22}) = \omega_\gamma J_2$.
- Stationary control space: $\mathcal{U}_0 = \{f \in L^2(0, 1) : \int f = 0\}$.

Appendix B: Spectral structure of K_0 and the restricted operator

The operator K_0 is compact and skew-adjoint, so $-K_0^2$ is positive, compact, and self-adjoint. Its largest eigenvalue s^2 equals $\|K_0\|^2$. Moreover, in the distributional or weak

sense,

$$(K_0 f)''' = 2f,$$

so the eigenvalue problem may be rewritten as an ODE system with boundary conditions. We define the exact constants by operator norms and do not use decimal values in the proof.

For the stationary projection, let $k_0(t, s) = \frac{1}{2}(t - s)|t - s|$ and

$$m(t) = \int_0^1 k_0(t, s) ds = \frac{t^3 - (1 - t)^3}{6}.$$

Then $P_0 K_0 P_0$ has kernel

$$k_{\text{stat}}(t, s) = k_0(t, s) - m(t) + m(s). \quad (\text{B1})$$

This kernel is antisymmetric and square integrable, so K_{stat} is compact and skew-adjoint on $\mathcal{U}_0$. Also $(K_{\text{stat}} f)''' = 2f$, hence $K_{\text{stat}} \neq 0$. Therefore the largest positive eigenvalue of $-K_{\text{stat}}^2$ exists, and its square root is $s_{\text{stat}} = \|K_{\text{stat}}\|$.

For Gauss–Legendre nodes t_i and weights w_i , the Nyström matrix is

$$M_{ij} = \sqrt{w_i} \frac{1}{2} (t_i - t_j) |t_i - t_j| \sqrt{w_j}.$$

Let $v_i = \sqrt{w_i}$ be the unit vector representing the constant function and set $P = I - vv^T$. The largest singular value of PMP approximates $\|K_{\text{stat}}\|$.

Appendix C: Details of exact recovery and realization by stationary waveforms

The unrestricted recovery construction in Theorem 4 lifts the quadratic optimizer to a ray in the exact endpoint map. Products of an odd number of STF control matrices return to the symmetric subspace, so the cubic term in the antisymmetric endpoint channel vanishes and the next term after quadratic order is quartic. Along $A_\rho = \rho A_*$, the cost is exactly $Q = 2\rho^2$ while

$$\omega = 2s_0\rho^2 + O(\rho^4).$$

The endpoint map is analytic in ρ , and therefore

$$\frac{d\omega}{d\rho} = 4s_0\rho + O(\rho^3) > 0$$

for sufficiently small $\rho > 0$. Exact small positive targets follow from one-sided monotonicity rather than the ordinary inverse-function theorem at $\rho = 0$, where the derivative vanishes. Reversing the sign of β_* yields the negative branch.

The stationary problem has the same structure, except that the optimizer is taken from the top singular pair of K_{stat} . Starting from a real unit eigenvector

$$-K_{\text{stat}}^2 \beta_* = s_{\text{stat}}^2 \beta_*,$$

set $\alpha_* = K_{\text{stat}} \beta_* / s_{\text{stat}}$. Since $\alpha_*, \beta_* \in \mathcal{U}_0$, the ray ρA_* preserves the tangent-space stationarity constraints for all amplitudes. The reconstructed strain

$$p_\rho(t) = 2\rho \int_0^t (t-s) \alpha_*(s) ds, \quad q_\rho(t) = 2\rho \int_0^t (t-s) \beta_*(s) ds$$

satisfies

$$p_\rho(0) = q_\rho(0) = p'_\rho(0) = q'_\rho(0) = p'_\rho(1) = q'_\rho(1) = 0.$$

Thus it is a physically admissible tangent waveform with an initially zero reference, stationary in/out behavior, and an arbitrary allowed constant final shear. This closes the admissibility step required to transfer sharpness from the unrestricted class to the physical stationary class.

Appendix D: Bel–Robinson normalization

We fix the explicit $1/4$ convention in Eq. (8). In vacuum, the null-screen Weyl tidal data reduce to two real degrees of freedom. With $m = (e_1 + ie_2)/\sqrt{2}$ and $T = \alpha\sigma_3 + \beta\sigma_1$, the Newman–Penrose scalar $\Psi_0 = C(k, m, k, m)$ agrees, up to the overall sign convention, with $\alpha + i\beta$. Hence

$$|\Psi_0|^2 = \alpha^2 + \beta^2 = \frac{1}{2} \text{tr}(T^2).$$

Under the normalization in Eq. (8), the fourfold null contraction is $B(k, k, k, k) = |\Psi_0|^2$, so

$$B(k, k, k, k) = |\Psi_0|^2 = \frac{1}{2} \text{tr}(T^2).$$

This gives Eqs. (10) and (11). Other normalizations of the Bel–Robinson tensor differ by numerical factors, so the explicit factor $1/4$ must be kept fixed when comparing formulas.

Appendix E: Perturbative order and order of limits

The Bondi/Jacobi relation is a leading radiation-zone statement. Let $C_{AB} = \varepsilon C_{AB}^{(1)} + O(\varepsilon^2)$ and $h_{AB}^{TT} = -C_{AB}/r + O(r^{-2})$. Our notation $X^{(2)}$ denotes the quadratic perturbative

coefficient in

$$X(\varepsilon) = X^{(0)} + \varepsilon X^{(1)} + \varepsilon^2 X^{(2)} + O(\varepsilon^3).$$

In particular,

$$Q_\gamma(\varepsilon) = \varepsilon^2 Q_\gamma^{(2)} + O(\varepsilon^3), \quad \omega_\gamma(\varepsilon) = \varepsilon^2 \omega_\gamma^{(2)} + O(\varepsilon^3), \quad \Delta\Psi_H(\varepsilon) = \varepsilon^2 \Delta\Psi_H^{(2)} + O(\varepsilon^3).$$

The inner ε^{-2} limit in Eq. (27) extracts these quadratic coefficients before the limit $r \rightarrow \infty$ is taken. We therefore do not claim a finite- r , finite-amplitude theorem in which the $O(\varepsilon^3)$ and $O(r^{-3})$ remainders are controlled simultaneously by one uniform constant.

Appendix F: Reproducibility ledger and stationarity audit

We distinguish analytic proof from floating-point validation. The numerical calculations check that the unrestricted K_0 optimizer does not automatically satisfy the mean-zero stationarity constraints, the Nyström convergence of $P_0 K_0 P_0$, and the exact-Jacobi recovery at small amplitude using the top singular pair. The sharp coefficient itself follows analytically from compact-operator theory; the decimal value 71.13876223... is used only as numerical validation.

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- [1] E. E. Flanagan, A. M. Grant, A. I. Harte, and D. A. Nichols, “Persistent gravitational wave observables: General framework,” *Phys. Rev. D* 99, 084044 (2019), arXiv:1901.00021, DOI: 10.1103/PhysRevD.99.084044.
 - [2] E. E. Flanagan, A. M. Grant, A. I. Harte, and D. A. Nichols, “Persistent gravitational wave observables: Nonlinear plane wave spacetimes,” *Phys. Rev. D* 101, 104033 (2020), arXiv:1912.13449, DOI: 10.1103/PhysRevD.101.104033.
 - [3] A. M. Grant and D. A. Nichols, “Persistent gravitational wave observables: Curve deviation in asymptotically flat spacetimes,” *Phys. Rev. D* 105, 024056 (2022), arXiv:2109.03832, DOI: 10.1103/PhysRevD.105.024056; Erratum *Phys. Rev. D* 107, 109902 (2023).
 - [4] A. Seraj and B. Oblak, “Gyroscopic Gravitational Memory,” *JHEP* 11, 057 (2023), arXiv:2112.04535, DOI: 10.1007/JHEP11(2023)057.
 - [5] A. Seraj and T. Neogi, “Memory effects from holonomies,” *Phys. Rev. D* 107, 104034 (2023), arXiv:2206.14110, DOI: 10.1103/PhysRevD.107.104034.

- [6] G. Faye and A. Seraj, “Gyroscopic Gravitational Memory from quasi-circular binary systems,” *Class. Quantum Grav.* 42, 035005 (2025), arXiv:2409.02624, DOI: 10.1088/1361-6382/ada339.
- [7] A. I. Harte, T. B. Mieling, M. A. Oancea, and E. Steininger, “Gravitational wave memory and its effects on particles and fields,” *Phys. Rev. D* 111, 024034 (2025), arXiv:2407.00174, DOI: 10.1103/PhysRevD.111.024034.
- [8] J. J. Ferrando and J. A. Sáez, “On the Bel radiative gravitational fields,” *Class. Quantum Grav.* 29, 075012 (2012), arXiv:1111.2755, DOI: 10.1088/0264-9381/29/7/075012.
- [9] S. Hahmann, B. Sauvage, and G.-P. Bonneau, “Area preserving deformation of multiresolution curves,” *Comput. Aided Geom. Des.* 22, 349–367 (2005), DOI: 10.1016/j.cagd.2005.01.006.
- [10] V. Ferone, B. Kawohl, and C. Nitsch, “The elastica problem under area constraint,” *Math. Ann.* 365, 987–1015 (2016), arXiv:1411.6100, DOI: 10.1007/s00208-015-1284-y.